
Algorithm 4 Offline Causal Matrix Completion

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1: Input: Multi-metric matrix  $X \in \mathbb{R}^{M \times T}$ , SCS  $\mathcal{G} = \{\mathcal{G}_1, \dots, \mathcal{G}_K\}$ ,
   adjacency matrix  $W \in \mathbb{R}^{M \times M}$ , causal orderings  $\pi = \{\pi_k\}_{1 \leq k \leq K}$ .
2: Output: Reconstructed matrix  $\hat{X}$ , per-DAG subspaces  $U = \{U_k\}_{1 \leq k \leq K}$ .
3: Initialise:  $\hat{X} \leftarrow 0$ ;  $U_k \leftarrow \emptyset$  for  $1 \leq k \leq K$ .
4: Draw random sample positions  $\Omega$  of size  $d = c\mu\tau \log(Kl/\delta) \leq T$ .
5: for each DAG  $\mathcal{G}_k = \langle \pi_k, W(\pi_k, \pi_k) \rangle$  do
6:   for each metric  $t \in \pi_k$  (in causal order) do
7:      $\mathbf{x}_t \leftarrow X(t, :)$ .
8:      $Pa(t) \leftarrow \{i \mid W(t, i) \neq 0\}$ .
9:      $U_{Pa(t)} \leftarrow \hat{X}(Pa(t), :)$ .
10:     $\mathbf{w}_t \leftarrow \mathcal{P}_\Omega(\mathbf{x}_t) \mathcal{P}_\Omega(U_{Pa(t)})^\dagger$ .
11:    if  $\|\mathcal{P}_\Omega(\mathbf{x}_t) - \mathbf{w}_t \mathcal{P}_\Omega(U_{Pa(t)})\|_2^2 > \eta$  then
12:       $\hat{X}(t, :) \leftarrow \mathbf{x}_t$ . ▷ Root: fully sample
13:       $U_k \leftarrow U_k \cup \{\mathbf{x}_t\}$ .
14:      Redraw  $\Omega$  of size  $d$ .
15:    else
16:       $\hat{X}(t, :) \leftarrow \mathbf{w}_t U_{Pa(t)}$ . ▷ Downstream: infer
17:    end if
18:  end for
19: end for

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A PROOF OF THE CAUSAL SAMPLING BOUND

This appendix provides the complete proof of Theorem 5.2. We first present an offline Causal Matrix Completion procedure (Algorithm 4) that demonstrates how, for a matrix $X \in \mathbb{R}^{M \times T}$, sampling only d elements per row suffices to recover the full matrix by exploiting the sparse causal structure (SCS). The proof then proceeds through three supporting results—a local rank characterisation (Lemma A.1), a rank-preservation guarantee under sampling (Proposition A.2), and a per-DAG recovery guarantee (Lemma A.3)—before assembling them into the main theorem via a union bound over the SCS.

We begin by establishing that each metric’s local neighbourhood has rank determined by its parents.

LEMMA A.1 (LOCAL RANK VIA CONDITIONAL INDEPENDENCE). *For any downstream metric m with parent set $Pa(m)$, consider the submatrix $B = X(\{Pa(m), m\}, :) \in \mathbb{R}^{(\tau_m+1) \times T}$. Under the conditional independence of parent metrics given their ancestral metrics, the SVD rank of B equals τ_m .*

PROOF. By the causal ordering produced by Algorithm 1, each parent metric $i \in Pa(m)$ has had the influence of its own ancestors removed. Concretely, after the deflation step (line 20 of Algorithm 1), each parent’s residual reduces to its exogenous noise term ϵ_i . Since the noise vectors $\{\epsilon_i\}_{i \in Pa(m)}$ are drawn from non-degenerate, mutually independent, non-Gaussian distributions (a standard assumption in linear causal models [68]), they are linearly independent with probability 1.

By the SEM (Equation 1), the target metric satisfies $\mathbf{x}_m = \sum_{i \in Pa(m)} w_{mi} \mathbf{x}_i + \epsilon_m$, so $\mathbf{x}_m$ lies in the span of its τ_m parent rows plus a noise component. Because $\mathbf{x}_m$ is a linear combination of the parents, the row space of B has dimension exactly τ_m : the τ_m parent rows are linearly

independent, and the row corresponding to m is linearly dependent on them. Hence $\text{rank}(B) = \tau_m$. $\square$

Next, we show that column sampling preserves this local rank structure.

PROPOSITION A.2 (RANK PRESERVATION UNDER SAMPLING). *Let $B = X(\{Pa(m), m\}, :) \in \mathbb{R}^{(\tau_m+1) \times T}$ be a rank- τ_m matrix with SVD $B = U\Sigma V^\top$, where $U \in \mathbb{R}^{(\tau_m+1) \times \tau_m}$, $\Sigma \in \mathbb{R}^{\tau_m \times \tau_m}$, $V \in \mathbb{R}^{T \times \tau_m}$. Let μ denote the incoherence of V , and let $B_{:, \Omega}$ be the submatrix formed by sampling columns of B with i.i.d. $\text{Ber}(d/T)$. If $d \geq 8\mu\tau_m \log(\tau_m/\delta)$, then with probability at least $1 - \delta$:*

$$\text{rank}(B_{:, \Omega}) = \tau_m.$$

PROOF. Let $Z = V^\top \in \mathbb{R}^{\tau_m \times T}$, which has orthonormal rows. Define the sampling matrix $S = \sum_{i=1}^T \delta_i Z_{:,i} \mathbf{e}_i^\top \in \mathbb{R}^{\tau_m \times T}$, where $\delta_i \sim \text{Ber}(d/T)$ and $\mathbf{e}_i$ is the i -th standard basis vector. Consider the positive semidefinite matrix $Y = SS^\top = \sum_{i=1}^T \delta_i Z_{:,i} Z_{:,i}^\top$.

Since Z has orthonormal rows, we have $\mathbb{E}[Y] = (d/T) ZZ^\top = (d/T) I_{\tau_m}$, giving $\mu_{\tau_m} = \lambda_{\tau_m}(\mathbb{E}[Y]) = d/T$. For each summand $Y_i = \delta_i Z_{:,i} Z_{:,i}^\top$, we have $\lambda_{\max}(Y_i) = \|\delta_i Z_{:,i}\|_2^2 \leq \|Z\|_{2,\infty}^2 \triangleq L$, where $\|Z\|_{2,\infty} = \max_{1 \leq i \leq T} \|Z_{:,i}\|_2$.

Applying the Matrix Chernoff Bound (Lemma A.4) with $\epsilon = 1/2$:

$$\begin{aligned} \Pr[\sigma_{\tau_m}(S) > 0] &= \Pr[\lambda_{\tau_m}(Y) > 0] \\ &\geq \Pr[\lambda_{\tau_m}(Y) > \tfrac{1}{2}\mu_{\tau_m}] \\ &\geq 1 - \tau_m \exp\left(-\frac{d \sigma_{\tau_m}^2(Z)}{8T\|Z\|_{2,\infty}^2}\right). \end{aligned}$$

Since $\sigma_{\tau_m}(Z) = \sigma_{\tau_m}(V^\top) = 1$ (orthonormal rows), the exponent simplifies. Setting this probability to $1 - \delta$ and solving for d yields the requirement $d \geq 8T\|Z\|_{2,\infty}^2 \log(\tau_m/\delta)$. By the incoherence property, $\|Z\|_{2,\infty}^2 \leq \max_{i \in [T]} \|V^\top \mathbf{e}_i\|_2^2 \leq \mu_{\tau_m}/T$, so the condition becomes $d \geq 8\mu\tau_m \log(\tau_m/\delta)$. When this holds, $\text{rank}(B_{:, \Omega}) = \text{rank}(Z_{:, \Omega}) = \tau_m$ with probability at least $1 - \delta$. $\square$

We now show that Algorithm 4 correctly identifies and recovers metrics within each DAG.

LEMMA A.3 (PER-DAG RECOVERY). *Consider the k -th DAG $\mathcal{G}_k = \langle \pi_k, W(\pi_k, \pi_k) \rangle$ with root metrics Root_k of size l_k . Let $U_k = X(\text{Root}_k, :) \in \mathbb{R}^{l_k \times T}$ denote the subspace spanned by root metrics. Suppose sample positions Ω are drawn with $d \geq c\mu\tau_t \log(\tau_t/\delta)$. For each metric $t \in \pi_k$:*

- (1) **Root detection.** *If $\mathbf{x}_t \in U_k$ but $\mathbf{x}_t \notin U_{Pa(t)}$ (i.e., t is a root metric or $\mathbf{x}_t$ cannot be represented by its parents), then $\text{rank}(\mathcal{P}_\Omega([U_{Pa(t)}; \mathbf{x}_t])) = \tau_t + 1$ with probability at least $1 - \delta$.*
- (2) **Downstream recovery.** *If $\mathbf{x}_t \in U_{Pa(t)}$ (i.e., t is a downstream metric), then $\text{rank}(\mathcal{P}_\Omega([U_{Pa(t)}; \mathbf{x}_t])) = \tau_t$ with probability 1, and the metric is exactly recovered via*

$$\mathcal{P}_\Omega(\mathbf{x}_t) \mathcal{P}_\Omega(U_{Pa(t)})^\dagger U_{Pa(t)} = \mathbf{x}_t \tag{30}$$

with probability at least $1 - \delta$.

- (3) **Outlier detection.** *If $\mathbf{x}_t \notin U_k$ (i.e., t is an outlier), then $\mathcal{P}_\Omega(\mathbf{x}_t)$ cannot be represented by its parent subspace with probability 1.*

PROOF. *Item 1.* Since $\mathbf{x}_t \notin U_{Pa(t)}$, we have $\text{rank}([U_{Pa(t)}; \mathbf{x}_t]) = \tau_t + 1$. By Proposition A.2, column sampling with $d \geq c\mu\tau_t \log(\tau_t/\delta)$ preserves this rank with probability at least $1 - \delta$, so

$$\text{rank}(\mathcal{P}_\Omega([U_{Pa(t)}; \mathbf{x}_t])) = \tau_t + 1$$

Item 2. Since $\mathbf{x}_t \in U_{Pa(t)}$, we have $\mathcal{P}_\Omega(\mathbf{x}_t) \in \text{rowspan}(\mathcal{P}_\Omega(U_{Pa(t)}))$ with probability 1, giving $\text{rank}(\mathcal{P}_\Omega([U_{Pa(t)}; \mathbf{x}_t])) = \tau_t$. For the recovery, let $\mathbf{w}_t = \mathcal{P}_\Omega(\mathbf{x}_t) \mathcal{P}_\Omega(U_{Pa(t)})^\dagger$ denote the estimated causal weight. The pseudo-inverse exists with probability at least $1 - \delta$ because Proposition A.2 ensures $\text{rank}(\mathcal{P}_\Omega(U_{Pa(t)})) = \tau_t$. By the SEM, $\mathbf{x}_t = \mathbf{w}_t U_{Pa(t)}$, and therefore $\mathbf{w}_t U_{Pa(t)} = \mathcal{P}_\Omega(\mathbf{x}_t) \mathcal{P}_\Omega(U_{Pa(t)})^\dagger U_{Pa(t)} = \mathbf{x}_t$.

Item 3. If $\mathbf{x}_t \notin U_k$, then $\mathbf{x}_t$ is drawn from a non-degenerate distribution independent of U_k , so $\text{rank}([U_k; \mathbf{x}_t]) = l_k + 1$ with probability 1, and no parent subspace can represent it. $\square$

We are now ready to prove the main theorem.

PROOF OF THEOREM 5.2. The proof combines the per-DAG guarantees of Lemma A.3 via a union bound over the SCS.

Item 1 (Recovery guarantee). Within the k -th DAG containing l_k root metrics, Item 1 of Lemma A.3 implies that root detection (which triggers full sampling) occurs at most l_k times. Each such event succeeds with probability $1 - \delta$, so by a union bound the joint success probability over all root detections is at least $1 - l_k\delta$. For downstream metrics (Item 2), recovery succeeds provided the pseudo-inverse exists, which depends on the same Ω that is redrawn at most l_k times. Applying a union bound over the l_k redraws, the recovery of all downstream metrics in the k -th DAG also succeeds with probability at least $1 - l_k\delta$. Combining both, all metrics in the k -th DAG are correctly recovered with probability at least $1 - 2l_k\delta$.

Across all K DAGs, let $l = \max_{1 \leq k \leq K} l_k$ and $\tau = \max_{1 \leq m \leq M} \tau_m$. Applying a union bound over the K DAGs, the joint success probability is at least $1 - 2Kl\delta$. To achieve a global guarantee of $1 - \delta$, we substitute $\delta/(2Kl)$ for δ in each per-metric bound, yielding the sample complexity $d \geq c\mu\tau \log(Kl/\delta)$ per row (absorbing the constant factor 2 into c).

Item 2 (Tighter bound). Let r_k denote the SVD rank of the k -th DAG's submatrix $X_i(\mathcal{V}_k, \cdot)$. Since each $X_i(\mathcal{V}_k, \cdot)$ is a submatrix of X_i , we have $r_k \leq r$. By Lemma A.1, the structurally relevant matrix $\mathbf{M}_m^G$ for any $m \in \mathcal{V}_k$ has $\text{rank } \tau_m \leq r_k$, so $\tau = \max_m \tau_m \leq \max_k r_k \leq r$. The overall comparison $c\mu\tau \log(Kl/\delta) \leq c\mu r \log(r/\delta)$ is dominated by $\tau \ll r$, since the logarithmic term grows slowly. When the SCS reflects a genuine sparse structure ($1 < K < M$), both τ and l are local quantities derived from individual DAGs and are significantly smaller than the global rank r . In degenerate cases— $K \rightarrow M$ (fully individual) or $K \rightarrow 1$ (fully joint)—the bound degenerates to $O(\tau \log M)$ or $O(\tau \log r)$ respectively, which remains no worse than $O(r \log r)$ since $\tau \leq r$. $\square$

For completeness, we state the matrix concentration inequality used in Proposition A.2.

LEMMA A.4 (MATRIX CHERNOFF BOUND [72]). Consider a finite sequence $\{\mathbf{X}_k\} \subset \mathbb{R}^{n \times n}$ of independent, random, Hermitian matrices satisfying $0 \leq \lambda_{\min}(\mathbf{X}_k) \leq \lambda_{\max}(\mathbf{X}_k) \leq L$. Define $\mathbf{Y} = \sum_k \mathbf{X}_k$ and let

$\mu_r = \lambda_r(\mathbb{E}[\mathbf{Y}])$ be the r -th largest eigenvalue of the expectation. Then for $\epsilon \in [0, 1)$:

$$\Pr[\lambda_r(\mathbf{Y}) > (1 - \epsilon)\mu_r] \geq 1 - r \left[\frac{e^{-\epsilon}}{(1 - \epsilon)^{1-\epsilon}} \right]^{\mu_r/L} \geq 1 - r e^{-\mu_r \epsilon^2/(2L)}.$$